



Regional Climate Responses to Geoengineering Interventions: Implications and Considerations for Climate Mitigation Strategies

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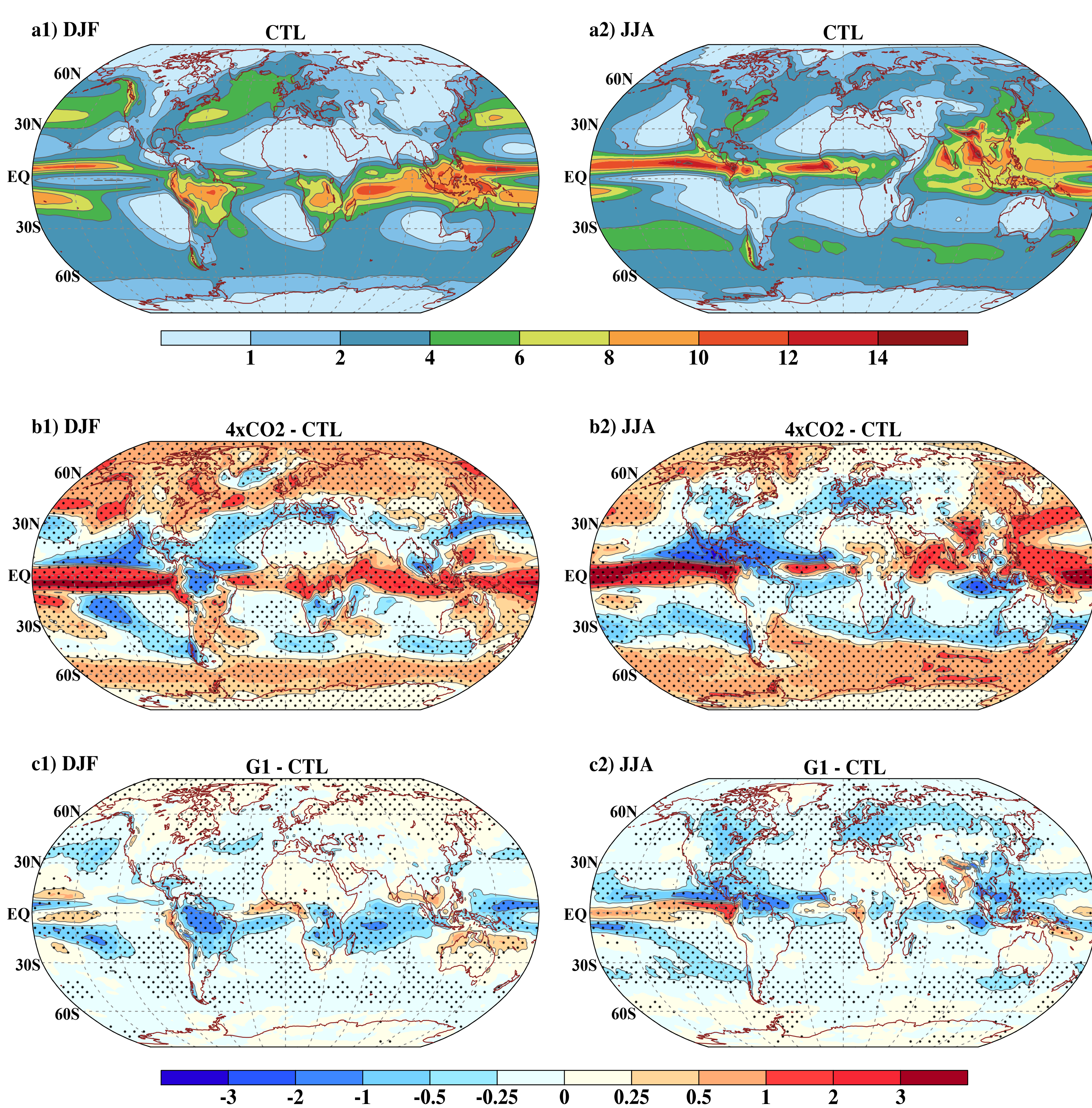
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Introduction

- Solar geoengineering, which comes forth as one of the promising ways to tackle global warming, albeit provisionally if not permanently, involves uncertainties in terms of perturbed regional climates with additional risks of unknown hazards.
- Our aim is to understand how circulation changes, particularly in regions strongly influenced by meridional temperature gradients, contribute to climate responses in geoengineered climates.
- We employed multi model mean of eight GeoMIP5 models – BNU-ESM, CESM1-CAM5, CCSM4, GISS-E2R, HadGEM2-ES, IPSL-CM5A-LR, MIROC-ESM and MPI-ESM-LR, assessing global precipitation and wind responses for 4xCO₂, G1 and, piControl (CTL) levels during boreal winter and summer.

Results and Conclusion

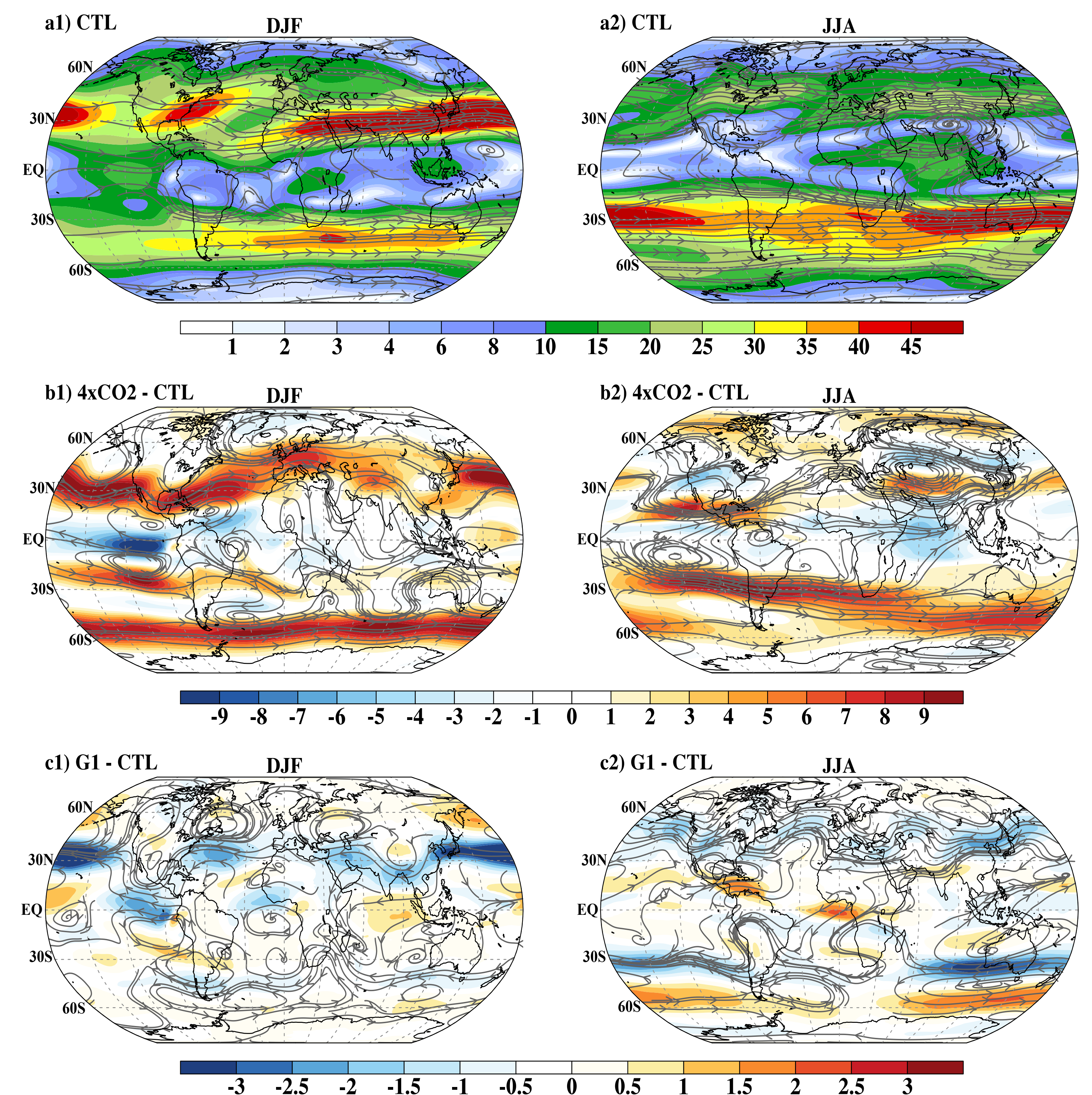


Spatial distribution of changes in precipitation in 4xCO₂ and G1 simulations w.r.t. CTL

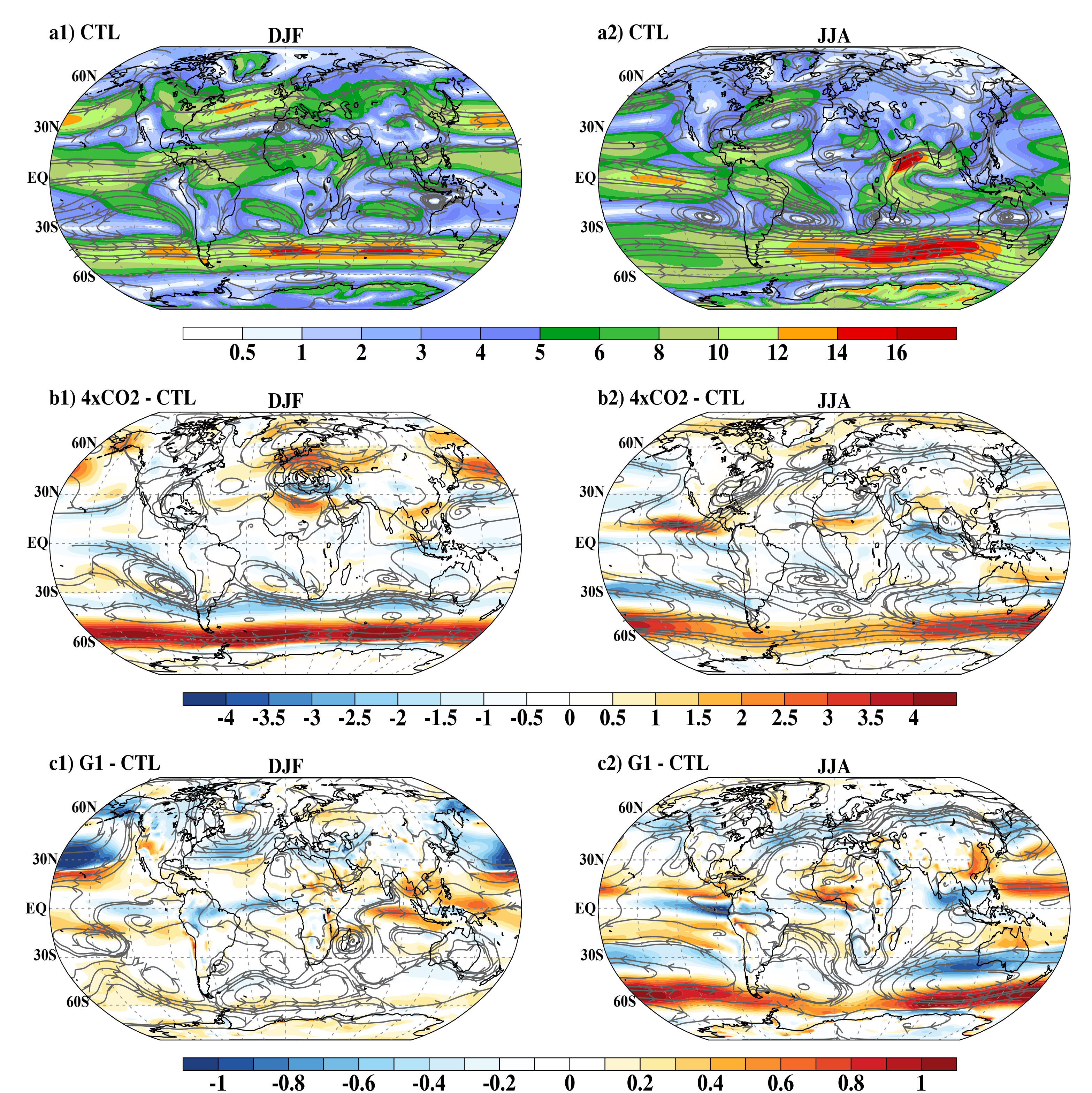
- Increased precipitation in deep tropics (> 2 mm/day) and sub-polar lows (0.5-1 mm/day) and decrease in the sub-tropical highs (0.25-1.0 mm/day) indicates strengthening of the atmospheric meridional circulation and the hydrological cycle in the low latitudes. The decrease in precipitation over Europe and most parts of the USA can be attributed to the dynamic changes in circulation at mid latitudes.
- The relocated Tibetan Anticyclone in a 4xCO₂ world has not completely moved to its observed location in the control climate and a partial anomalous anticyclonic motion persists over central Asia. The consistent weakening of the westerlies both at the lower and upper level over the latitudinal belt (30N-60N), can be vividly observed in a geoengineered world.
- In absence of a revolutionary transformation of energy systems throughout the world and the lesser degree of feasibility of such transformations, added with the ineffective societal response motivates the exploration of other strategies like solar geoengineering to mitigate global warming and climate change.

References

- [1] Tilmes et al. (2013). The hydrological impact of geoengineering in the Geoengineering Model Intercomparison Project (GeoMIP).
- [2] Robock, A. (2008). 20 reasons why geoengineering may be a bad idea.



Spatial distribution of changes in winds at 200 hPa in 4xCO₂ and G1 simulations w.r.t. CTL



Spatial distribution of changes in winds at 850 hPa in 4xCO₂ and G1 simulations w.r.t. CTL

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